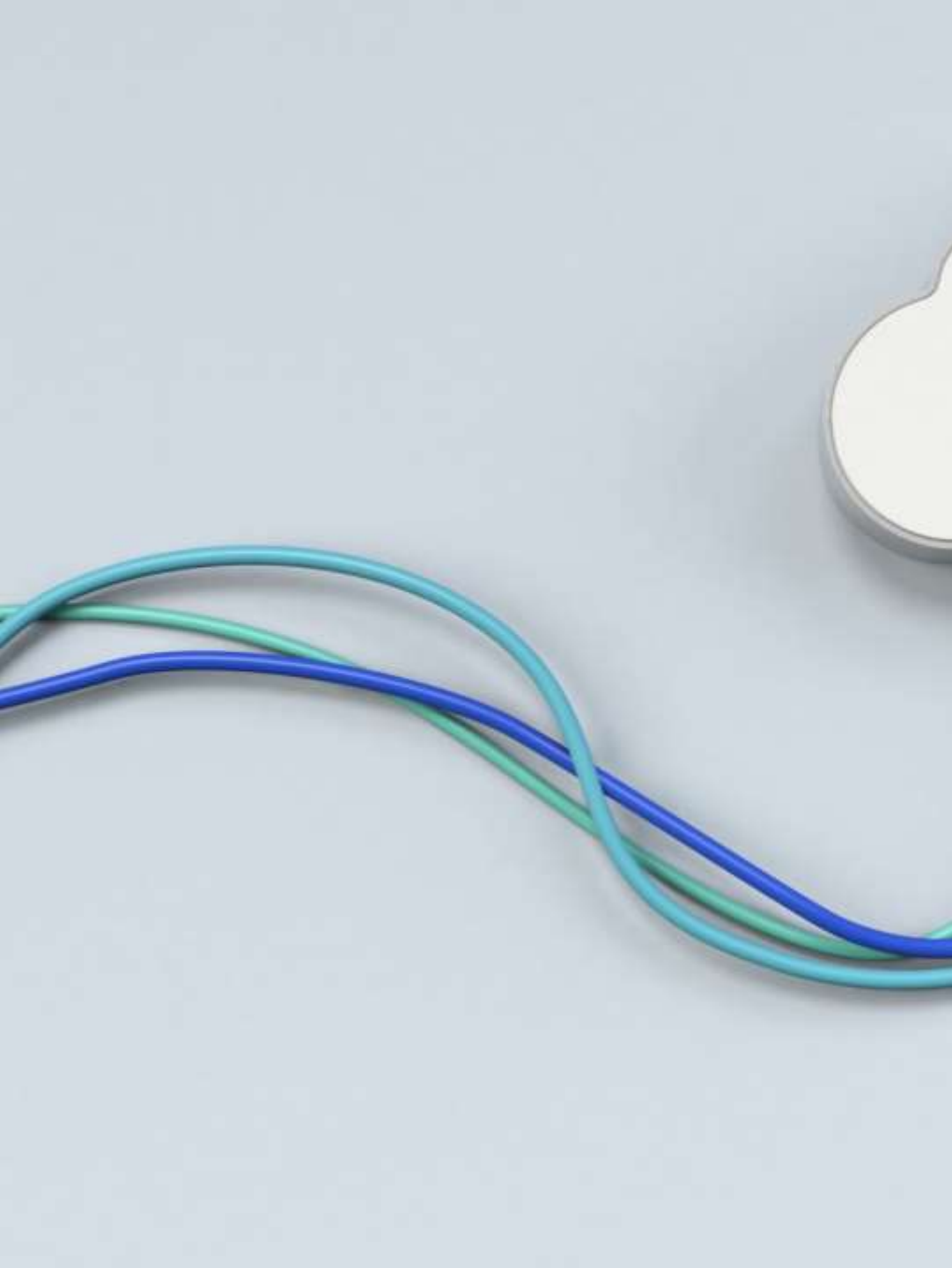




Topic 2

Architectural Layers of Cloud Computing



Contents

- Introduction to Cloud Computing
- Cloud Computing Layers
- Software as a Service (SaaS)
- Platform as a Service (PaaS)
- Infrastructure as a Service (IaaS)
- Dynamic Infrastructure
- Summary & Q&A

Cloud Computing?

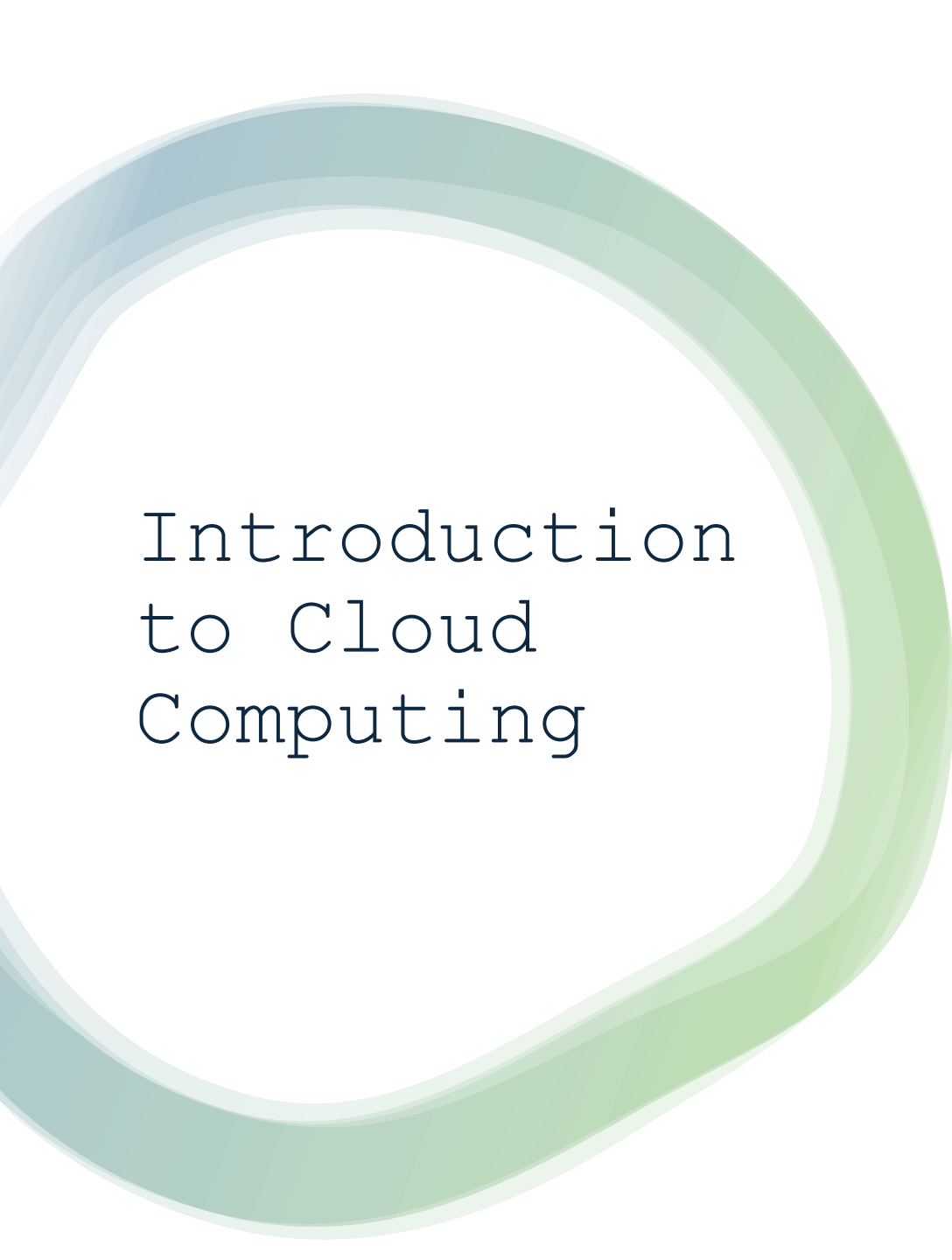


Computers can fly now?

quickmeme.com

IF I USE CLOUD COMPUTING

**WOULD I LOSE DATA DURING
RAIN?**



Introduction to Cloud Computing

- Cloud computing is the on-demand delivery of computing services (servers, storage, databases, networking, software) over the internet.
- **Key Characteristics:**
 - On-demand self-service
 - Broad network access
 - Resource pooling
 - Rapid elasticity
 - Measured service (pay-as-you-go)



Delivery of Computing Services

- Servers
- Storage
- Databases
- Networking
- Software
- Analytics
- Artificial Intelligence

Why Cloud Computing?



Business

Advantages:

- Scalability for growing workloads
- Cost-effective model (operational vs. capital expenses)
- Faster deployment of new services
- Global reach and availability



Technical

Advantages:

- Automated software updates
- Simplified infrastructure management
- Resilience and redundancy

Cloud Computing Layers

High-Level Layers in Cloud:

1. Infrastructure as a Service (IaaS)
 2. Platform as a Service (PaaS)
 3. Software as a Service (SaaS)
- **Goal:** Abstract complexity for end users and businesses.



Understanding the Layered Model

- **Layered Approach:**

- Each layer builds upon the capabilities of the layer below.
- Users at each layer only focus on resources they directly manage.

- **Analogy:**

- SaaS: The complete package (like using a fully-furnished apartment).
- PaaS: The foundation to build upon (like an empty apartment where you bring your own furniture).
- IaaS: The raw materials and space (like the land and building framework).



Software as a

Service (SaaS)
• Software distribution model in which applications are hosted by a third-party provider and made available over the internet.

- **Key Characteristics:**

- No local installation required
- Subscription or pay-per-use pricing
- Accessible via web browser or thin client

- **Examples:**

- Salesforce, Microsoft 365, Google Workspace (Gmail, Docs), Dropbox



SaaS – Benefits and Challenges



Benefits:

- Quick deployment and minimal setup
- Reduced IT overhead (maintenance, updates are handled by provider)
- Access from anywhere, on multiple devices



Challenges:

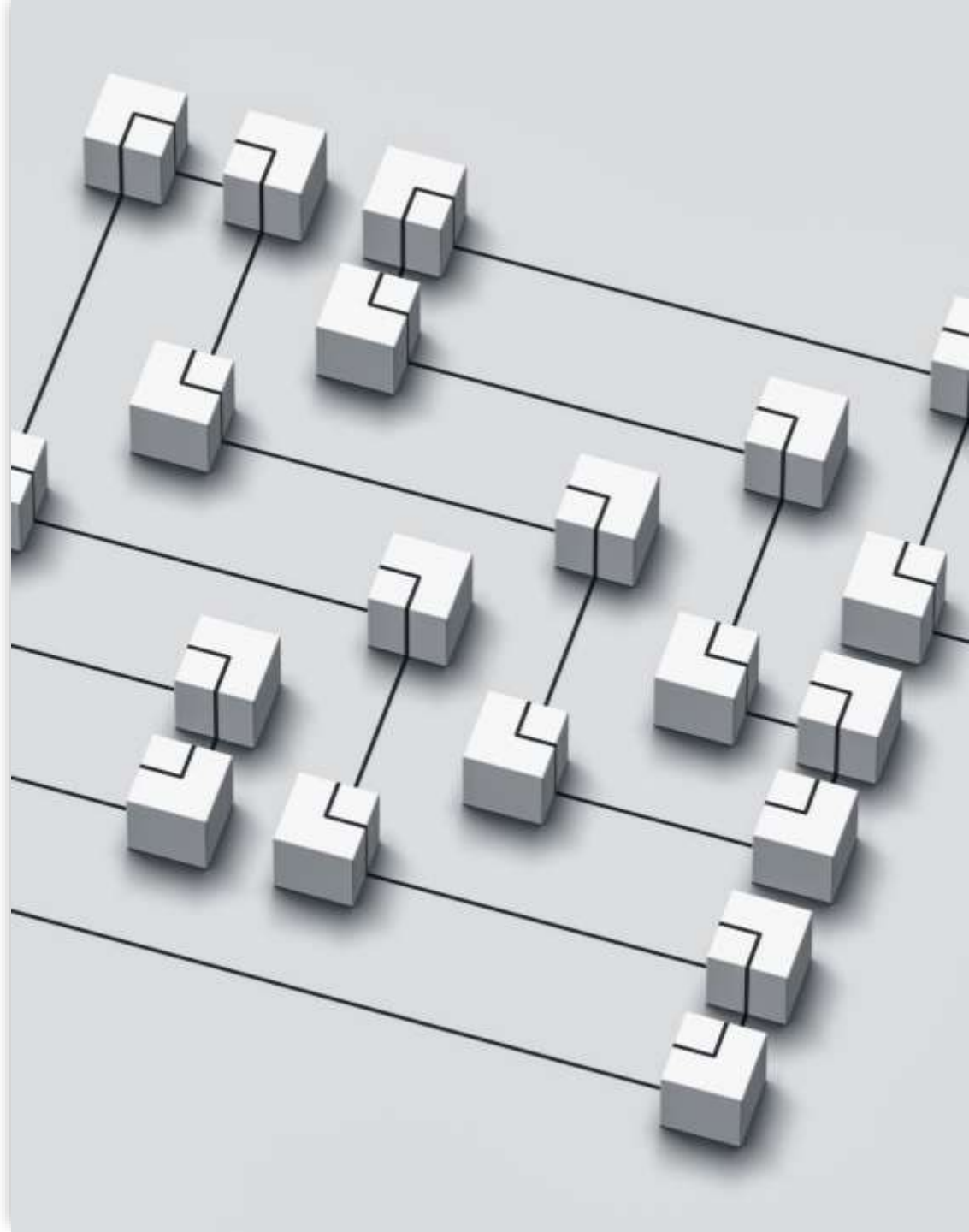
- Limited customization compared to on-premises software
- Dependency on internet connectivity
- Potential data privacy and compliance concerns



Software As A
Service (SAAS)

Platform as a Service (PaaS)

- A set of tools and services designed to make coding and deploying applications quick and efficient.
- Providers manage the underlying infrastructure; developers focus on the application itself.
- **Key Components:**
 - Runtime environment (language runtimes, frameworks)
 - Development tools
 - Databases and integration services



PaaS – Benefits and Challenges



Benefits:

- Simplifies development and deployment
- Less complexity in managing servers, storage, or networking
- Automated scalability



Challenges:

- Might be restricted to provider-specific tools/languages
- Risk of “vendor lock-in”
- Costs can grow if the application scales rapidly

Examples of PaaS Providers



Major PaaS Offerings:

Heroku (Salesforce)

Google App Engine (Google Cloud Platform)

Azure App Service (Microsoft Azure)

AWS Elastic Beanstalk (Amazon Web Services)



Key Services Offered:

Hosting environments

Integrated development environments (IDEs)

Database and storage integrations



AWS Elastic
Beanstalk



App Engine



docker



kubernetes

Platform As A Service (PAAS)



Infrastructure as a Service (IaaS)

- Virtualized computing resources over the internet, including servers, storage, and networking hardware.
- Users install and manage operating systems, applications, and middleware on provided infrastructure.
- **Key Components:**
 - Virtual machines (VMs)
 - Storage (object, block, file)
 - Networking (virtual networks, firewalls, load balancers)

IaaS – Benefits and Challenges

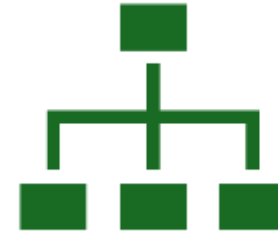


Benefits:

High flexibility and control over resources

Pay-as-you-go model for infrastructure

Easy scalability and rapid provisioning



Challenges:

Requires in-house expertise to configure and manage

Security and compliance are partly the customer's responsibility

Potential complexity in managing virtual machines and networks

IaaS Providers and Use Cases



Major IaaS Providers:

Amazon Web Services (EC2, S3, etc.)

Microsoft Azure (Virtual Machines, Azure Storage)

Google Cloud Platform (Compute Engine, Cloud Storage)



Typical Use Cases:

Hosting web applications

Big data processing

Backup and disaster recovery

Development and testing environments

Infrastructure As A Service (IAAS)



Google Cloud



DigitalOcean



Alibaba Cloud



Microsoft Azure

On-Premises

Applications

Data

Runtime

Middleware

o/s

Virtualization

Servers

Storage

Networking

Applications

Data

Runtime

Middleware

o/s

Virtualization

Servers

Storage

Networking

Infrastructure as a Service

- Applications
- Data
- Runtime
- Middleware
- o/s
- Virtualization
- Servers
- Storage
- Networking

Applications

Data

Runtime

Middleware

o/s

Virtualization

Servers

Storage

Networking

The diagram illustrates the PaaS architecture stack, showing the layers from the user applications down to the underlying hardware. The layers are represented as stacked rectangles, with the top two layers (Applications and Data) in light blue and the remaining layers (Runtime, Middleware, o/s, Virtualization, Servers, Storage, and Networking) in light orange. The stack is centered on a light gray background.

Platform as a Service

- Applications
- Data
- Runtime
- Middleware
- o/s
- Virtualization
- Servers
- Storage
- Networking

Applications

Data

Runtime

Middleware

o/s

Virtualization

Servers

Storage

Networking

```
graph BT; Applications --> Data; Data --> Runtime; Runtime --> Middleware; Middleware --> o/s; o/s --> Virtualization; Virtualization --> Servers; Servers --> Storage; Storage --> Networking
```

Software as a Service

Applications

Data

Runtime

Middleware

o/s

Virtualization

Servers

Storage

Networking

Applications

Data

Runtime

Middleware

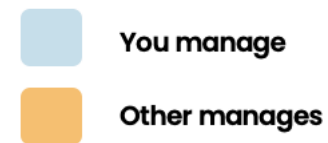
o/s

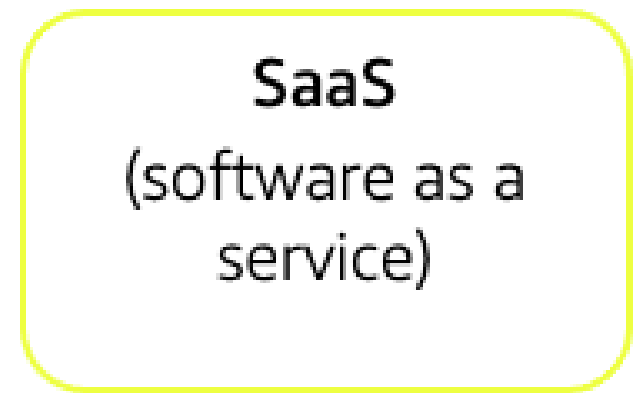
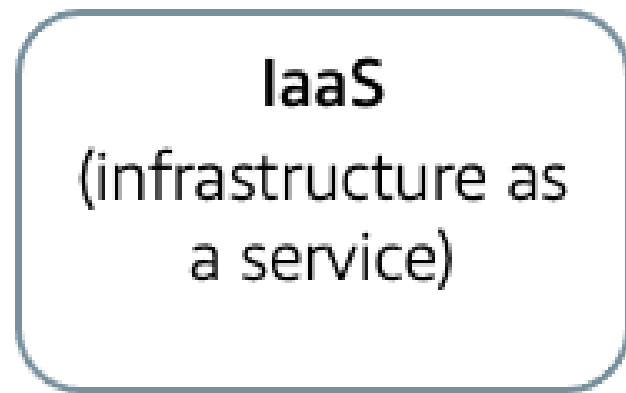
Virtualization

Servers

Storage

Networking





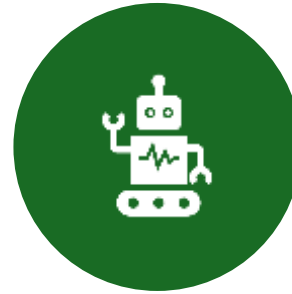
More control
over IT resources

Less control
over IT resources

Dynamic Infrastructure



The ability to quickly adapt and reconfigure IT infrastructure resources as demands change.



Often enabled by virtualization, automation, and orchestration technologies.



Ensures optimal performance, cost efficiency, and high availability



Facilitates continuous integration and continuous deployment (CI/CD)

Key Components of Dynamic Infrastructure



Virtualization -
Abstracts hardware into
virtual resources (VMs or
containers)



Automation - Scripts or
tools to streamline
provisioning,
configuration, and
scaling



Orchestration - Systems
(like Kubernetes, Docker
Swarm) that manage the
lifecycle of multiple
resources



Monitoring & Management -
Tools to observe and
adjust resource usage in
real time (e.g., AWS
CloudWatch, Azure
Monitor)

Benefits of a Dynamic Infrastructure

Scalability

- Easily add or remove resources in response to traffic or workload changes

Agility

- Faster deployment of new applications and updates

Cost Optimization

- Pay only for the resources actually in use

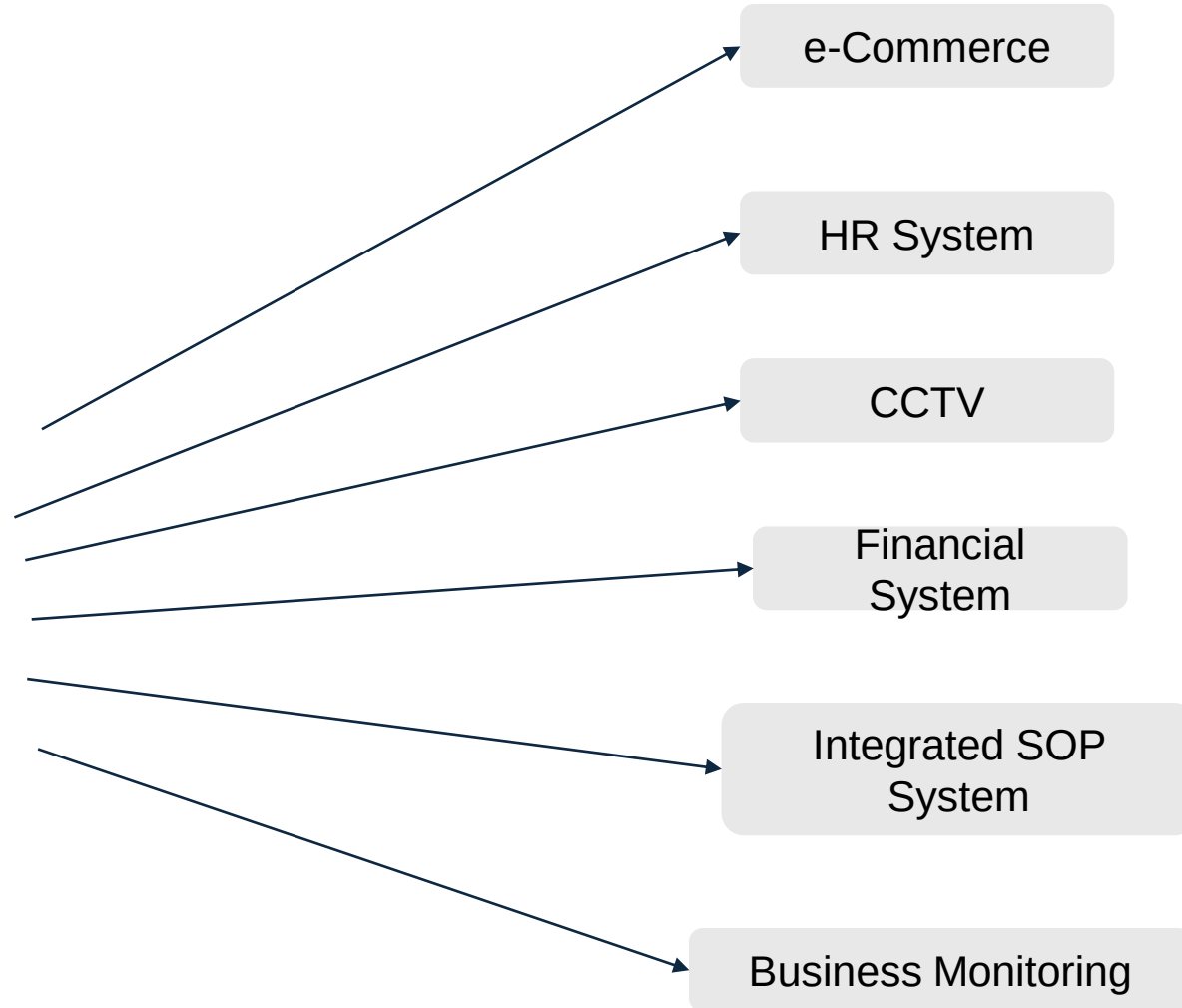
Fault Tolerance

- Automatically shift workloads if a server or region fails

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HQ & Izrulfizal
Shoe Store



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Shoe Store



Traditional way:

- Weeks/months to setup the infrastructure
- Running business on paper

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Shoe Store



Modern way:

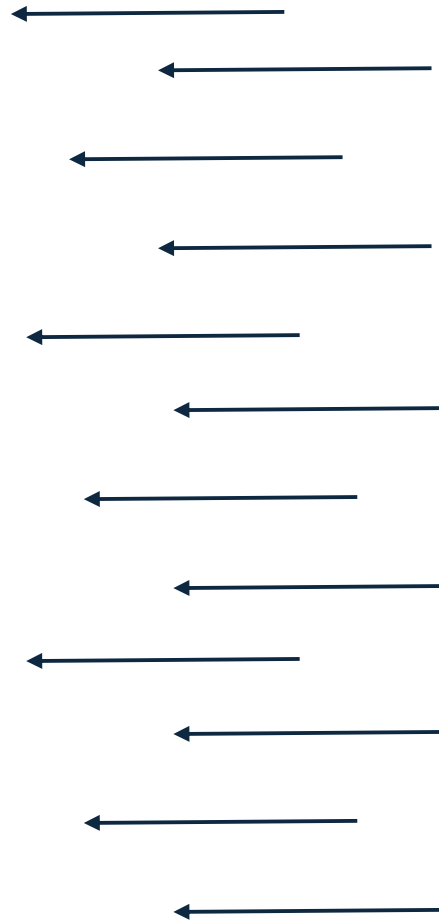
- Cloud hosting
- Server deployment in minutes
- Everything integrated
- Easy accessibility anytime

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Shoe Store

Heavy Traffics



Scalable

Vertical Scaling

(Increase size of instance (RAM ,
CPU etc.))



Horizontal Scaling

(Add more instances)



Challenges of Dynamic Infrastructure



Complexity

Multiple layers of abstraction can be difficult to manage



Skill Gap

Advanced knowledge required for automation/orchestration tools



Governance

Ensuring compliance and security policies in rapidly changing environments



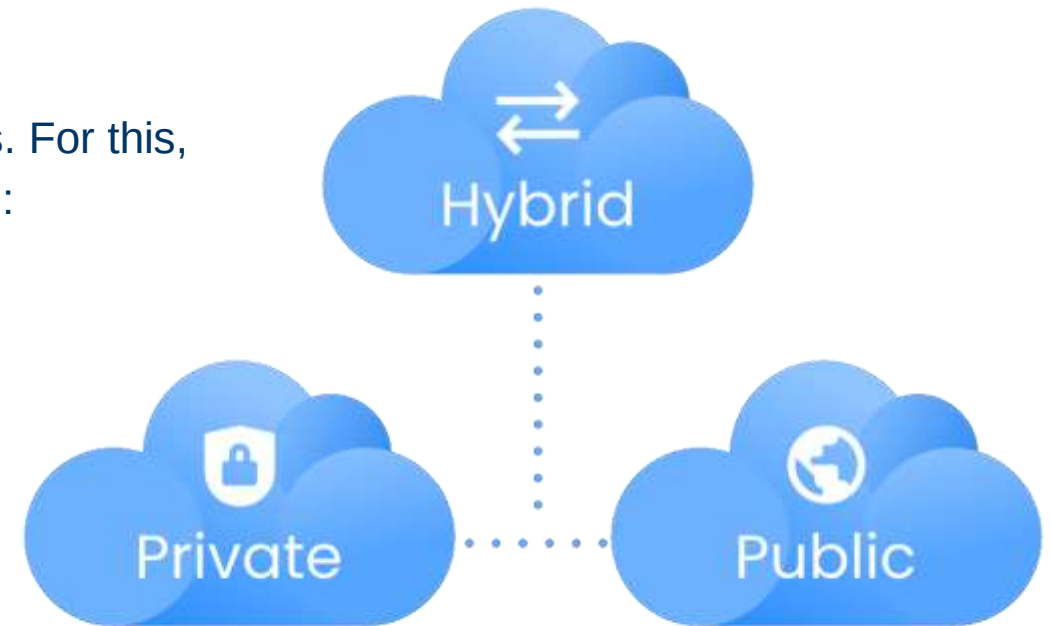
Monitoring & Debugging

Dynamic scaling can make troubleshooting more complex

Type of Cloud Deployment

Another thing about Cloud Computing is how accessible it is. For this, we can differentiate cloud computing in three different ways :

- **Public Cloud**
- **Private Cloud**
- **Hybrid Cloud -**
 - **Mixture of Private & Public**



Putting It All Together



Layered Approach Recap

SaaS: Complete software solutions provided to end users

PaaS: Platform that abstracts infrastructure and offers tools for development

IaaS: Core infrastructure (servers, storage, networking) delivered virtually



Dynamic Infrastructure -
Underpins all these layers with
flexibility and automation



Choosing the Right Model -
Depends on your business needs,
development expertise, and
desired level of control

Summary

- Cloud computing offers different service models (SaaS, PaaS, IaaS) to suit various needs.
- Dynamic infrastructure enables scalability, agility, and cost-effectiveness.
- Understand the trade-offs (control vs. complexity, cost vs. flexibility).

